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ME-504 Project

Predicting Von Mises Stress
Using CNNs: Regression vs
Classification (Q7)

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Problem Statement

Physical Setup

- Cantilever beam of centered rectangular unit cells
 - Each strut has a different thickness (226 structural nodes)
 - A point load is applied at the mid right-edge of the beam
 - FEM simulation run 5,000 times with random thickness profiles
 - Goal: Train a CNN to replace the expensive FEM solver

Q7 Objectives

- Approach 1 — CNN Regression
Predict the exact maximum Von Mises stress value
- Approach 2 — CNN Classification
Classify stress into 4 percentile buckets (0-10, 11-25, 26-50, 51-100)
- Compare both approaches: metrics, data requirements & practicality

Data Overview

output.xlsx : Input X

- 5,000 rows × 226 columns
- Each row = 1 thickness profile
- Each column = thickness at 1 node
- Values range: 0.1 → 0.9

stress/ folder : Target y

- 5,000 text files
- 3,304 rows × 3 columns each
- Column 1 = Von Mises stress
- We extract MAX value per file

cord.txt : Grid Shape

- 3,304 rows × 2 columns
- X & Y node coordinates
- Count unique X → grid Width
- Count unique Y → grid Height

Preprocessing Pipeline

01 StandardScaler on X

- Thickness values range 0.1 → 0.9 across 226 nodes
- Formula: $X_{\text{scaled}} = (X - \text{mean}) / \text{std}$
- Result: mean=0, std=1 per feature
- Why? Prevents CNN from ignoring low-range features

02 Log-transform + Normalize y

- Stress values span ~500,000 → 10,000,000 Pa (huge range!)
- Step A: $y_{\text{log}} = \log(y_{\text{raw}})$ → compresses to small numbers
- Step B: $y_{\text{norm}} = (y_{\text{log}} - \text{mean}) / \text{std}$ → centred around 0
- Eval: $\text{stress}_{\text{pred}} = \expm1(y_{\text{norm}_{\text{pred}}} \times \text{std} + \text{mean})$

03 Reshape X → 2D Spatial Map

- cord.txt gives grid dimensions (H × W)
- Reshape 226 values → (N, 1, H, W) tensor
- Now spatial neighbours are neighbours in the grid too
- This is why 2D CNN outperforms 1D CNN

Approach 1 : CNN Regression

Predict the exact maximum Von Mises stress value

Network Architecture

Input	(N, 1, H, W) — thickness spatial map
Conv Block 1	Conv2D(32, 3×3) → BN → ReLU → MaxPool
Conv Block 2	Conv2D(64, 3×3) → BN → ReLU → MaxPool
Conv Block 3	Conv2D(128, 3×3) → BN → ReLU → MaxPool
Flatten	Converts 2D feature maps to 1D vector
FC Layers	Dense(256) → Dropout(0.3) → Dense(64)
Output	Dense(1) — single scalar stress value

Loss Function

Huber Loss

Robust to outliers combines
L1 + L2 behaviour with threshold δ

Results

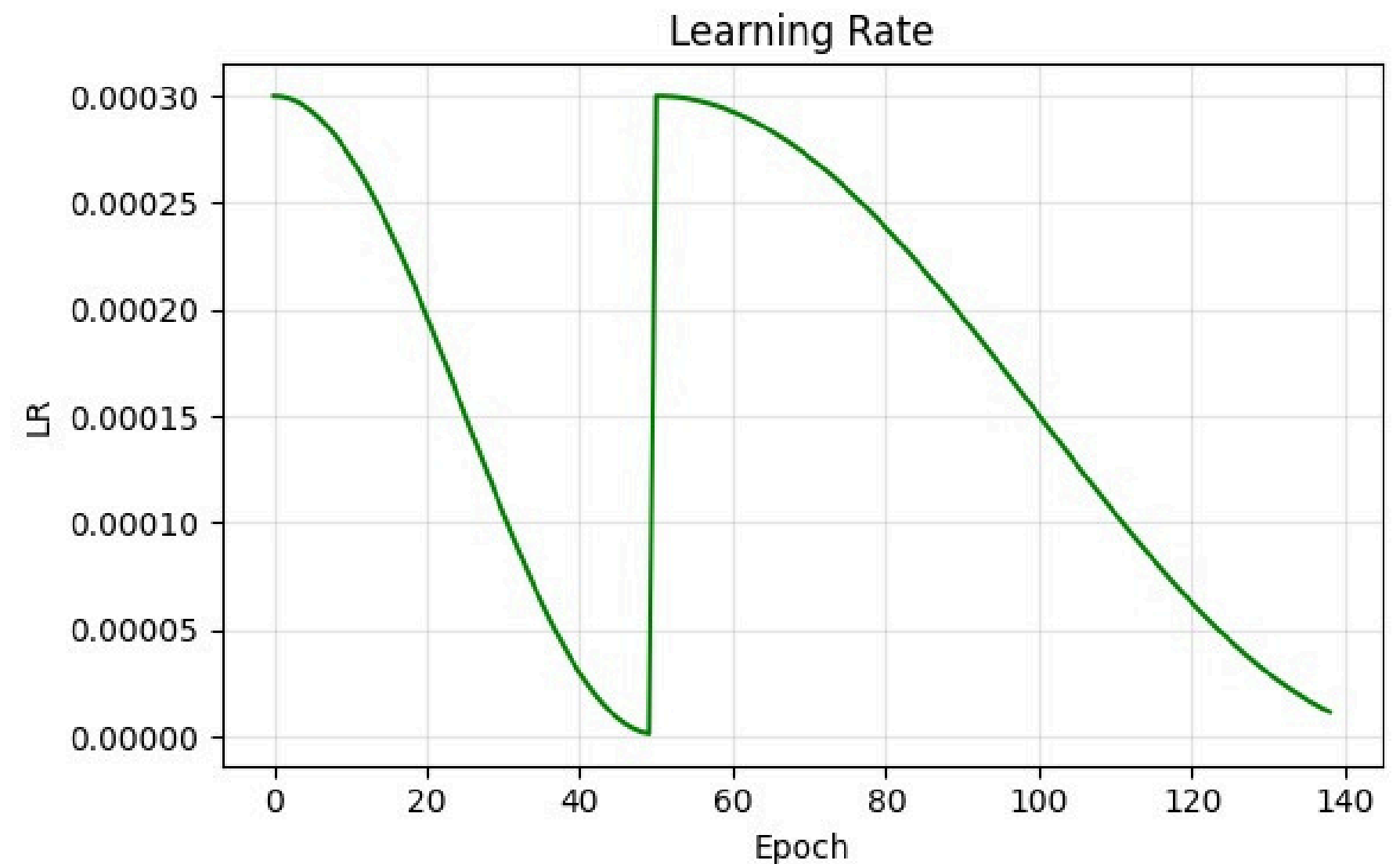
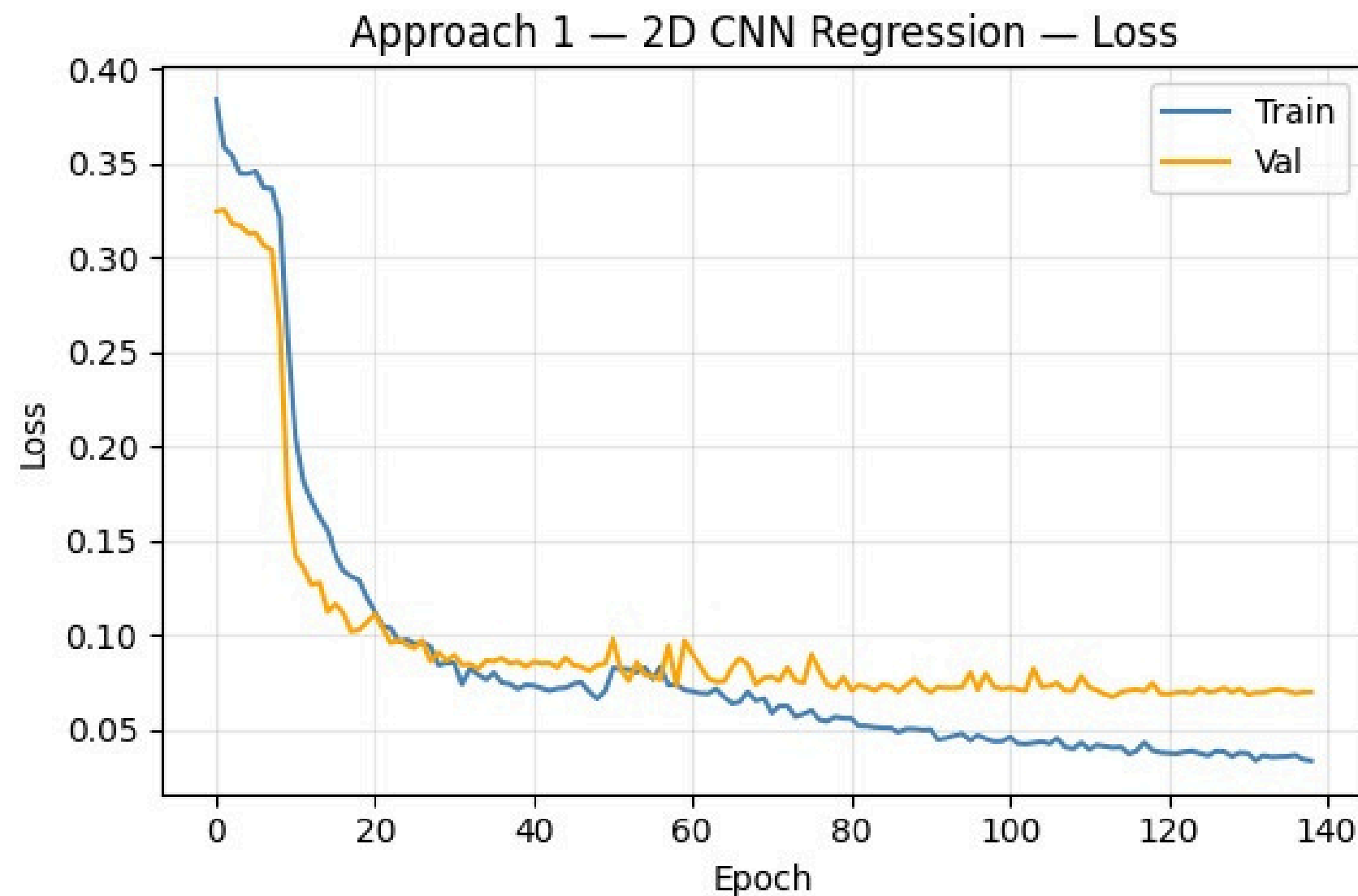
R² **0.836**

83.6% variance explained

MAPE **9.58%**

Mean abs % error

Approach 1 : Training Curves



Cosine Annealing LR

Learning rate starts at $3e-4$, decays smoothly via cosine schedule. It resets at epoch 50 (Warm Restarts CosineAnnealingWarmRestarts). This helps escape local minima.

Train vs Val Loss

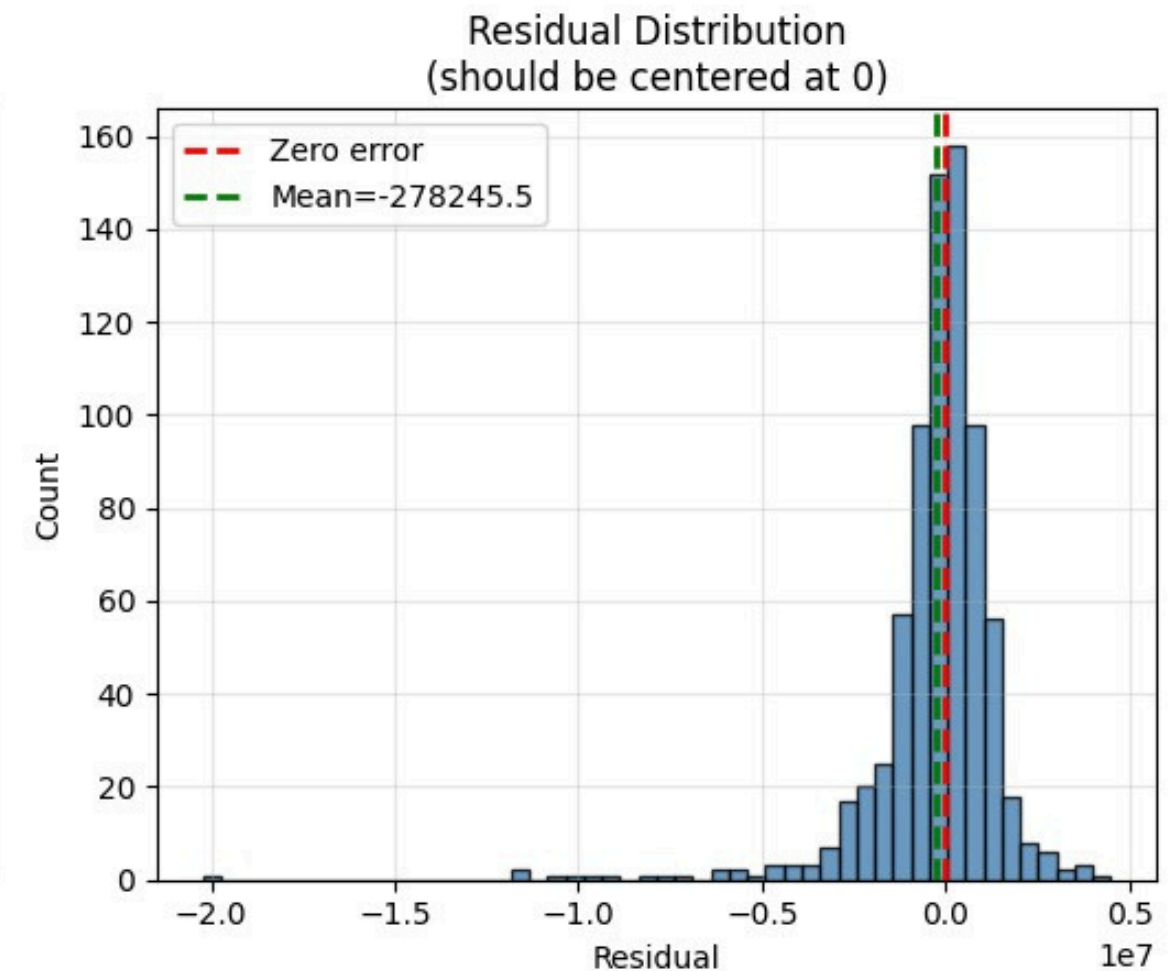
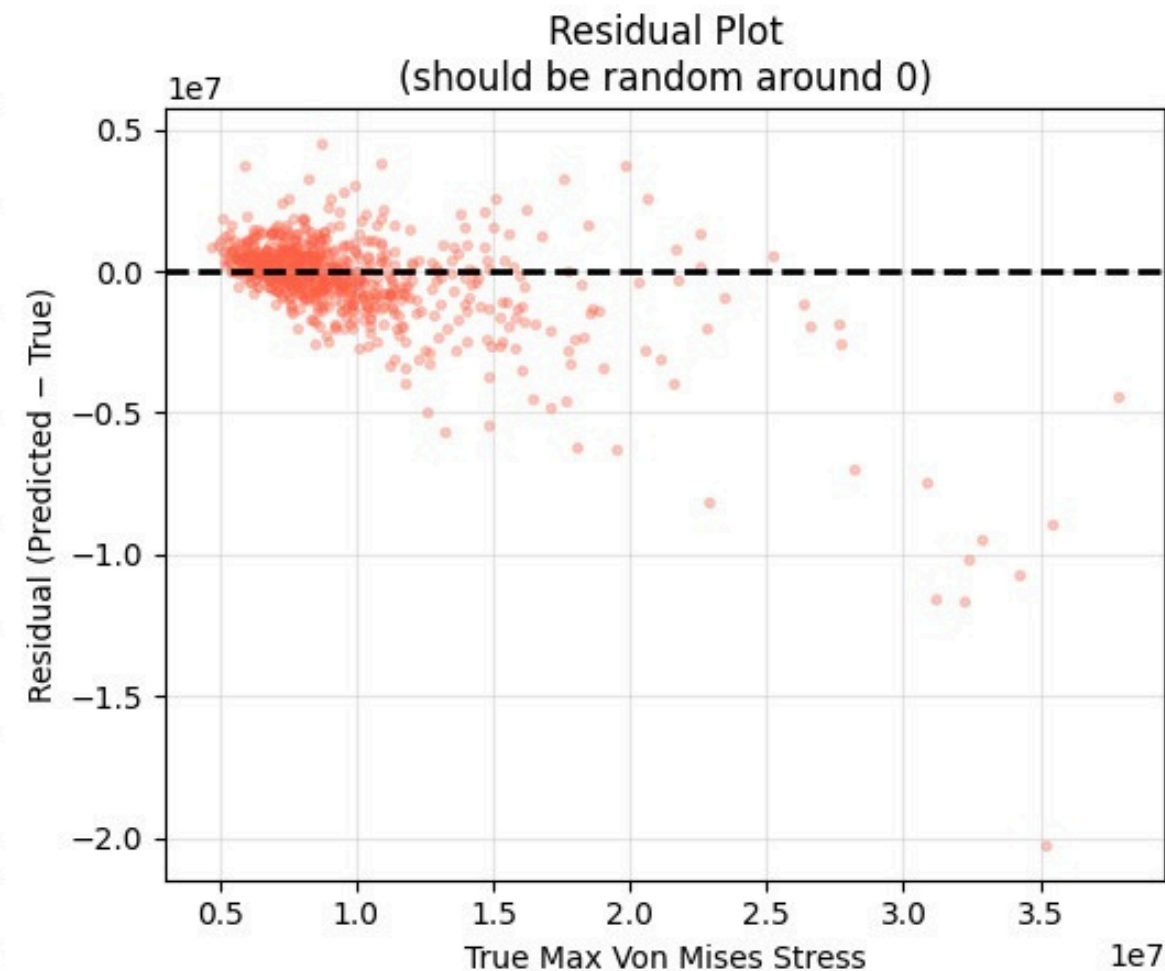
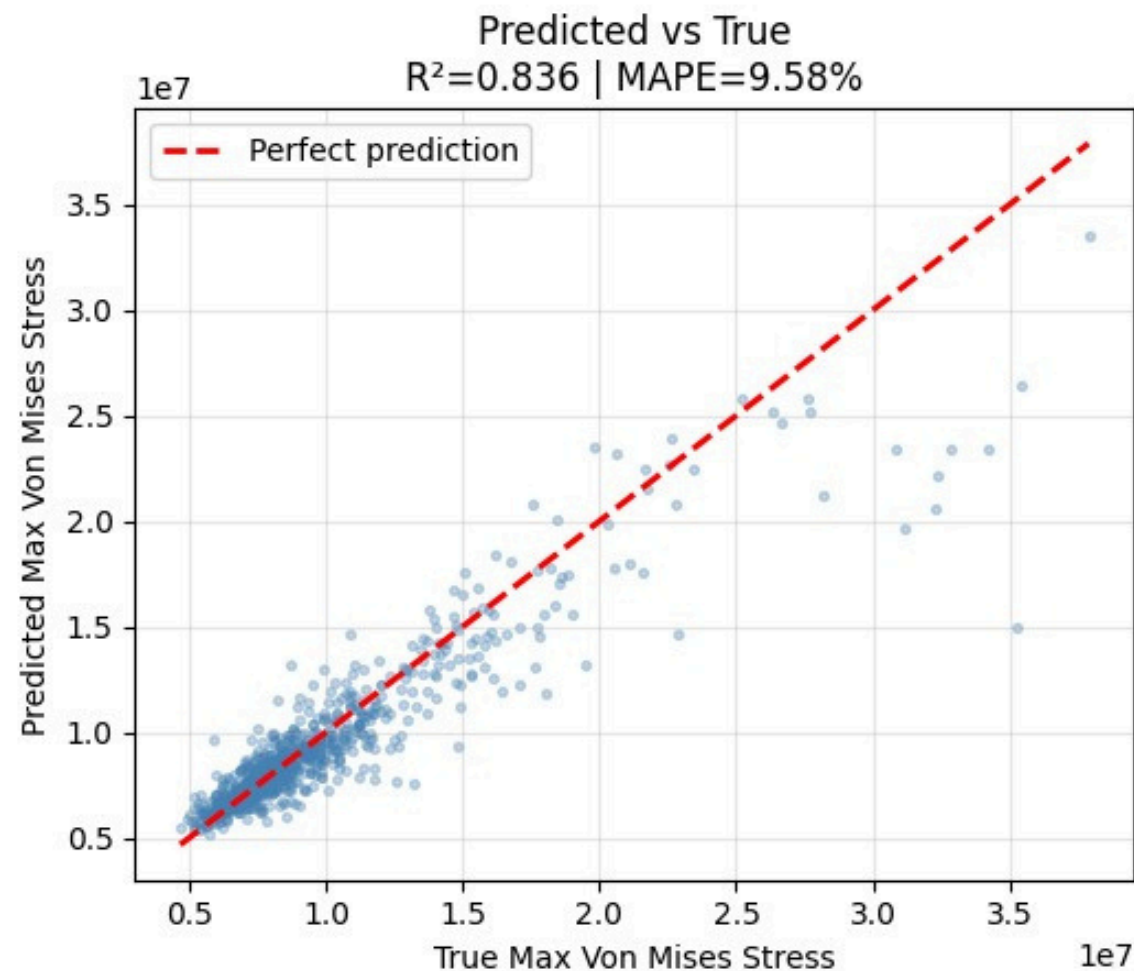
Both curves decrease together no over-fitting. Train loss goes lower (~ 0.04) while Val plateaus at ~ 0.07 . Gap is small, indicating good generalisation.

Convergence

Loss drops sharply in first 20 epochs then stabilises. The LR reset at epoch 50 causes a brief bump then further improvement a sign warm restarts are working.

Approach 1 : Evaluation Plots | $R^2=0.836$ | MAPE=9.58%

Approach 1 — 2D CNN Regression Evaluation



Predicted vs True (Left)

Points cluster tightly along the red dashed perfect-prediction line. $R^2=0.836$ means 83.6% of stress variance is explained. Scatter increases at higher stress harder cases.

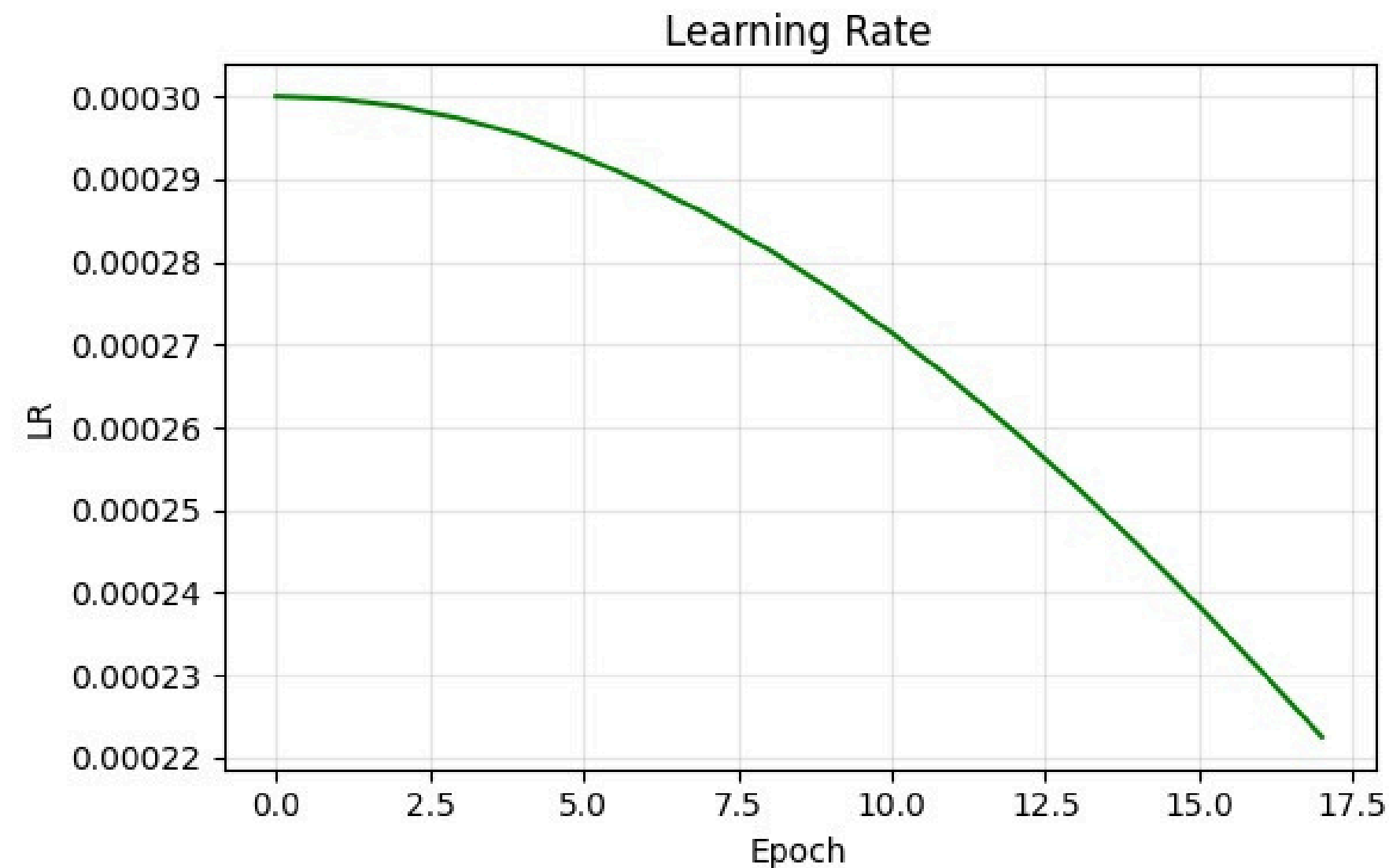
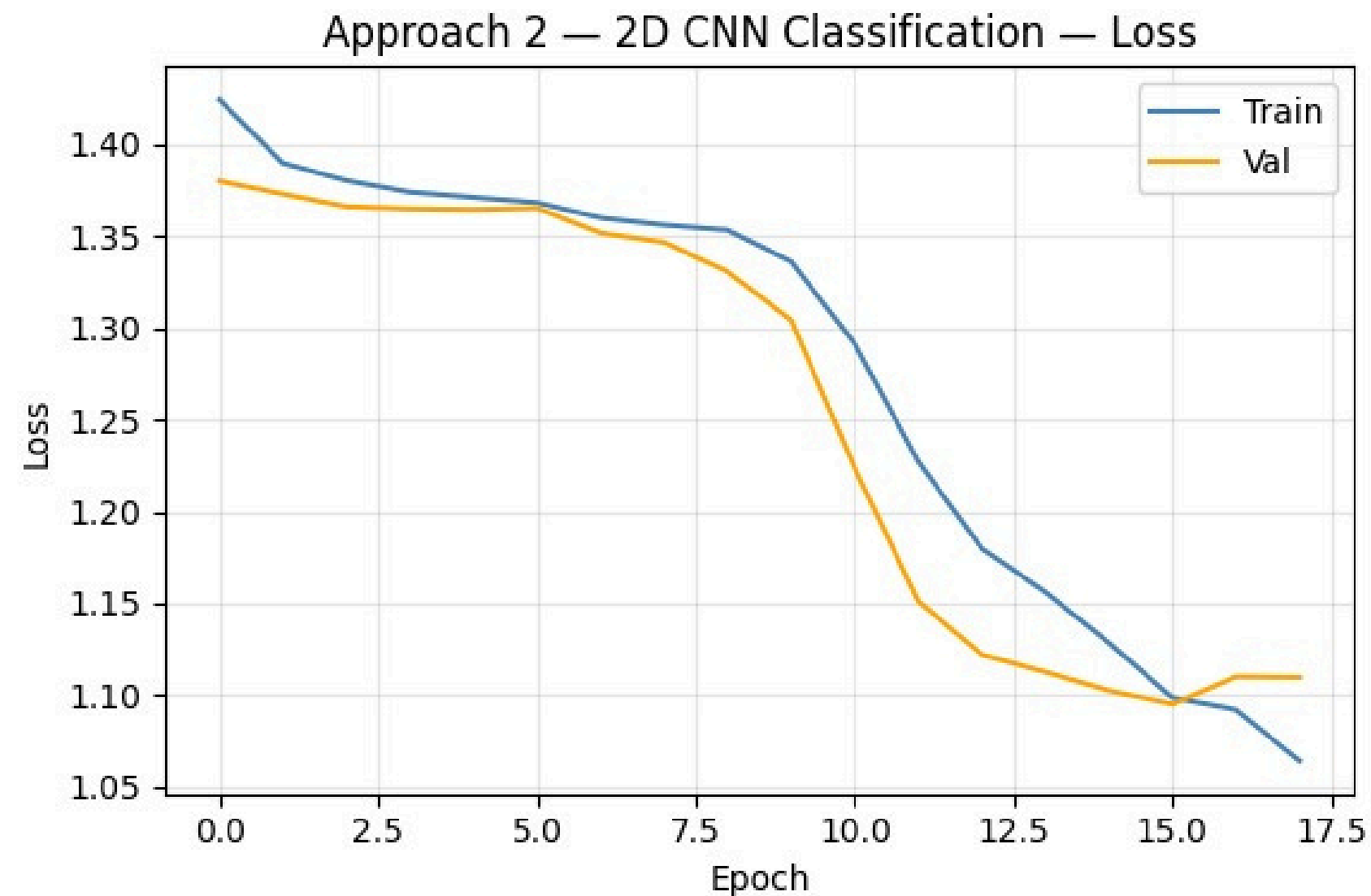
Residual Plot (Centre)

Residuals (Predicted - True) scatter around 0 (black dashed line). Mean = -278,245 Pa shows a slight under-prediction bias. No systematic pattern = well-behaved model.

Residual Distribution (Right)

Histogram is centred near zero and roughly bell-shaped. Slight left skew means the model occasionally under-predicts large stresses normal for regression on physical data.

Approach 2 : Training Curves



Faster Convergence

Classification converges in only 17 epochs vs 140 epochs for regression. Categorical labels are a simpler learning target than exact real-valued stress.

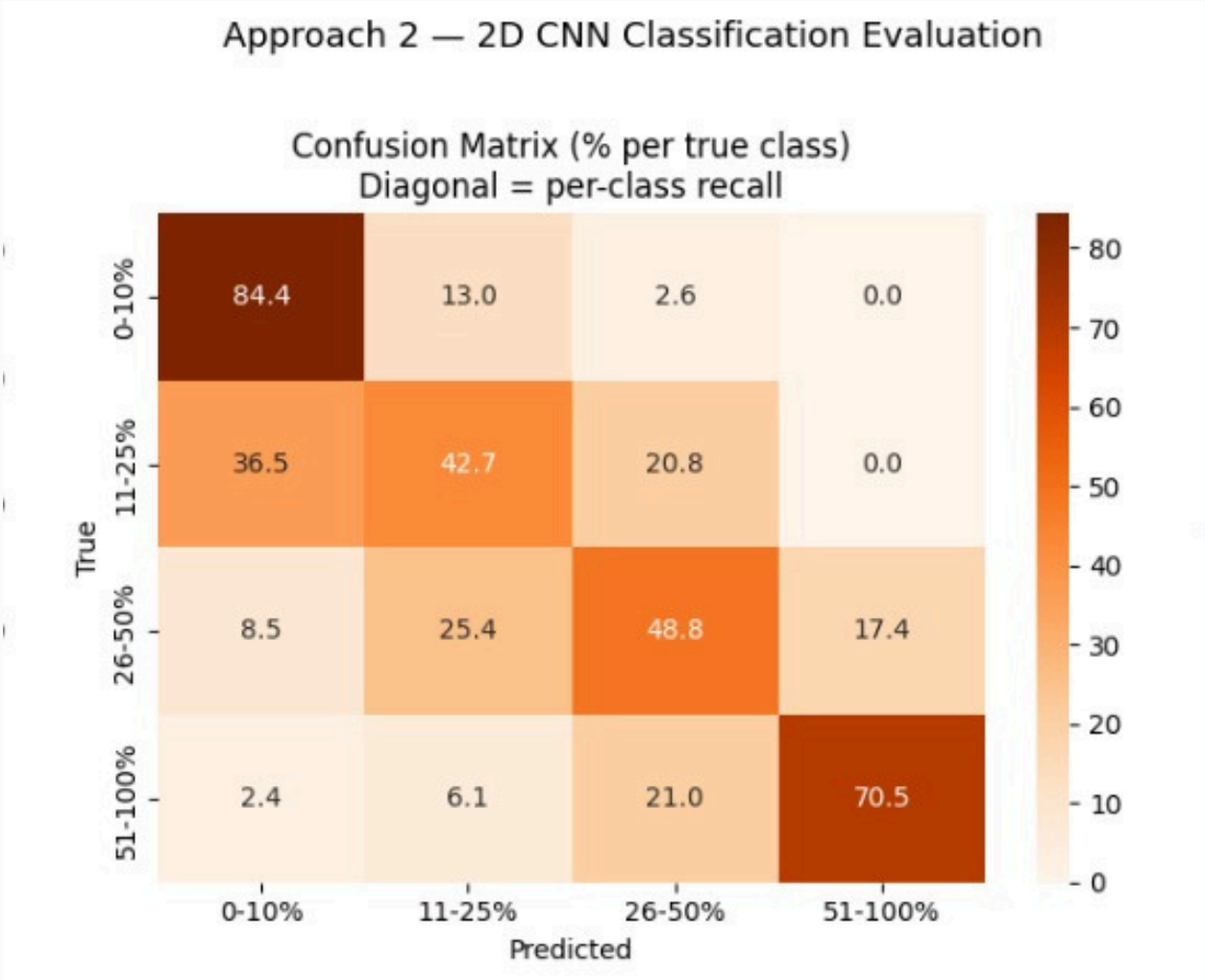
Val Loss < Train Loss

Validation loss is lower than training loss throughout, which can happen with label smoothing and dropout active only during training.

Smooth LR Decay

No LR restarts here (plain CosineAnnealing). LR decays from $3e-4 \rightarrow 2.2e-4$. Fewer epochs needed so warm restarts were not necessary.

Approach 2 : Confusion Matrix | Accuracy = 62.5%



Class 0 (0–10%ile)

84.4%

Best recall class weights successfully prioritised this rare class. Model identifies the safest designs very well.

Class 1 (11–25%ile)

42.7%

Most confused gets mixed with Class 0 (13%) and Class 2 (20.8%). Small sample size makes this hardest to learn.

Class 2 (26–50%ile)

48.8%

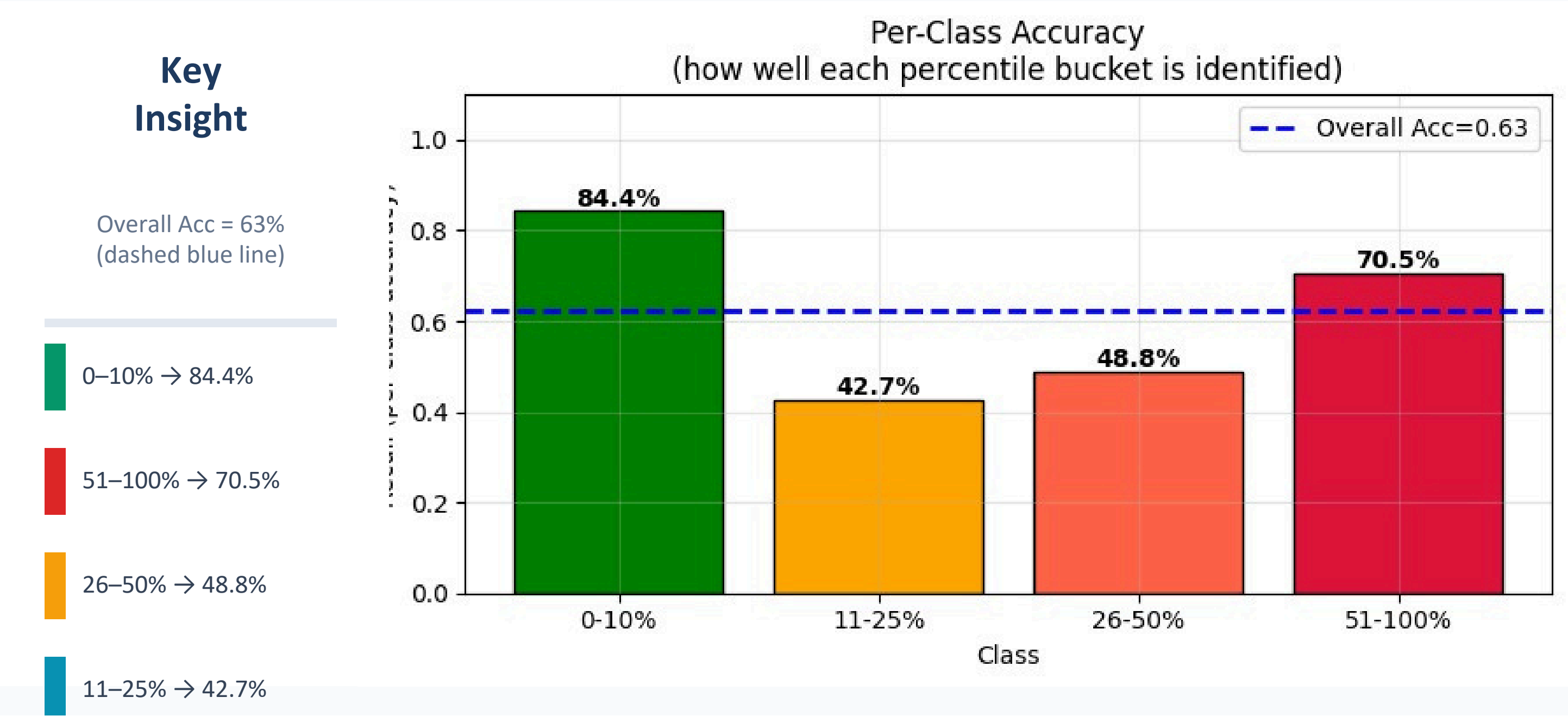
Moderate performance. Confusion mainly with Class 1 (25.4%) and Class 3 (17.4%) a natural boundary problem.

Class 3 (51–100%ile)

70.5%

Good recall despite having most data. Majority class still benefits from weighting not over-predicted anymore.

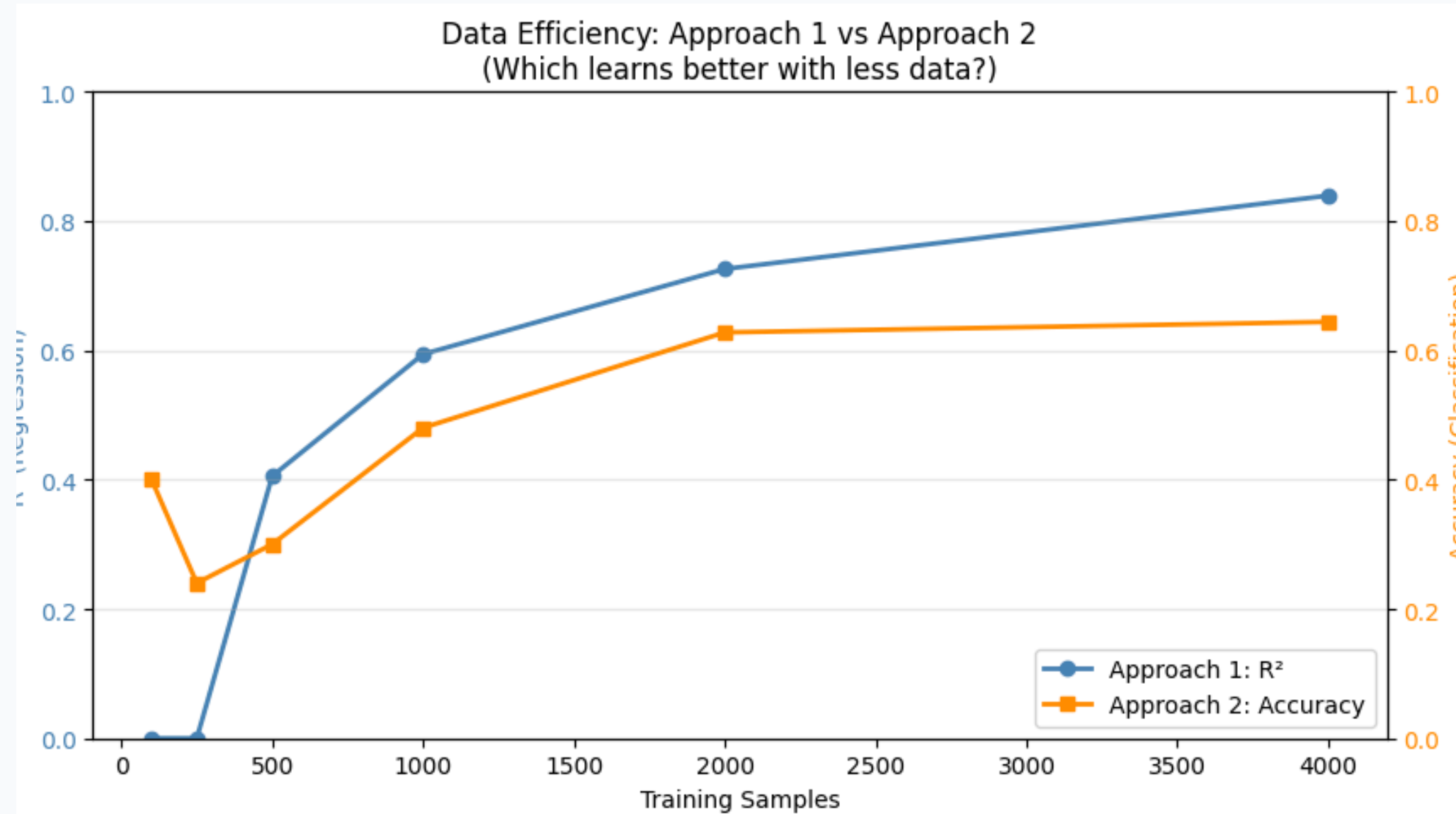
Approach 2 : Per-Class Accuracy



Why this pattern?

Class 0 (84.4%) is the easiest to identify designs with very low stress have distinctly thin/uniform thickness distributions. Class 3 (70.5%) is also well-identified since the most dangerous designs have clearly concentrated thicknesses in high-stress zones. Classes 1 & 2 (42–49%) are harder because they occupy the middle ground their thickness patterns overlap, making boundary decisions difficult. This is a fundamental challenge of percentile-based classification: adjacent classes share similar features.

Comparison of Approach 1 and Approach 2



APPROACH 1: Regression (predict exact stress value)

Needs MORE data must learn a precise numerical mapping across ALL stress values, not just relative ordering

APPROACH 2: Classification (predict percentile bucket)

Needs LESS data — relative ordering is easier to learn than absolute magnitudes in the millions (Pa scale)

VERDICT: Which is SUPERIOR?

- For ENGINEERING DECISIONS (need exact stress):
→ Approach 1 wins
- For RAPID SCREENING with limited data (<500 samples):
→ Approach 2 wins

Conclusions

- 01 2D CNN successfully learns thickness-to-stress mapping by treating the node grid as a spatial image unlike 1D which misses spatial relationships.
- 02 Regression ($R^2=0.836$, MAPE=9.58%) delivers exact stress values directly usable for structural optimisation and design decisions.
- 03 Classification (62.5% accuracy, 17 epochs) trains 8× faster, needs less data, and is ideal for rapidly screening thousands of candidate designs.
- 04 Three bugs (file sort, log-transform, class imbalance) each caused major performance failures. Systematic debugging improved R^2 from -0.007 to 0.836 .
- 05 Use Regression for exact engineering answers; use Classification for fast design-space exploration.

Thank You